

LinguaQuest

Overview

LinguaQuest is a gamified listening comprehension mobile application designed to help Grade 7 Filipino learners master seven key listening skills by sailing alongside Captain Salita and the student-sailor Dagat across the Listening Sea. Across seven themed islands, students collect Shards of Understanding to assemble the Alingawngaw Crystal and silence Ingay, the Spirit of Noise. Built with modern cross-platform development practices using Expo SDK 55, React Native 0.83, and a Next.js 16 + Prisma + Neon backend with real-time multiplayer powered by Pusher.

Features

Learning Modules

- **7 Story-Based Islands**, each tied to a Listening Skill
 - Island 1: Isla ng Salita — *Vocabulary in Context*
 - Island 2: Isla ng Bilis — *Listening at Speed / Pacing*
 - Island 3: Isla ng Diwa — *Main Idea & Theme*
 - Island 4: Isla ng Damdamin — *Tone, Mood & Emotion*
 - Island 5: Isla ng Tanong — *Specific Details & Facts*
 - Island 6: Isla ng Kwento — *Sequence & Narrative Flow*
 - Island 7: Isla ng Alingawngaw — *Full Application & Final Trial*
- **Pin-Based Challenge Progression** with multiple challenge pins and checkpoints per island
- **Audio Listening Passages** authored as full reading scripts and recorded for voice playback
- **Captain Salita Story Frame** weaving every island into one continuous quest narrative
- **Persistent Journey State** with shard collection and per-island progress tracking

Listening & Assessment System

- **Audio-Based Multiple-Choice Challenges** — listen to a passage, answer four-option (A/B/C/D) questions
- **Single-Play Audio Discipline** — passages play once to train focused, attentive listening

- **Per-Question Hint System** with thoughtful, skill-targeted hints written for every challenge
- **Captain-Read Explanations** — when a learner answers correctly, Captain Salita reads the explanation aloud through a dedicated explanation audio clip
- **Pass Threshold of 80%** per pin, mirroring a teacher-tuned mastery bar
- **Accuracy Tracking (0–100)** stored per pin for both student and teacher dashboards
- **Ingay Sabotage Encounters** — the Spirit of Noise taunts learners with a per-island warning before key challenges, reinforcing focus

Multiplayer Mode — *The Crew Voyage*

- **Real-Time Crew Rooms** with shareable join codes hosted via Pusher channels
- **5-Round Ship Repair Battles** — the crew defends a ship with five parts (hull, mast, sails, anchor, rudder), each starting at 25 HP
- **Crew Voting System** — players vote on the answer for every question; majority answer becomes the crew's response
- **Repair Voting Phase** between rounds where players choose which ship part to repair
- **Damage & Repair Logic** of ± 25 HP per round, driven by the crew's correct/incorrect answer
- **Question Timeout Handling** to keep the round moving even when players go idle
- **In-Room Crew Chat** with a dedicated ChatMessage channel and 300-character limit
- **Treasure Map View** showing the active round, ship status, and crew progress

Gamification & Rewards

- **Interactive Pirate Map** showing all seven islands with locked/unlocked progression states
- **Shard Collection System** — every completed island grants a unique Shard of Understanding tied to that skill
- **Badge & Achievement System** with milestone, mastery, and streak categories
- **Alingawngaw Crystal Assembly** — collecting all seven shards completes the meta-quest
- **Certificate of Completion** generated and exportable from the profile screen via react-native-view-shot

- **Animated Quest Scenes** including Shard Claim Cinematics and Quest Scene Overlays
- **Animated Captain Salita and Dagat Characters** that react to story beats

Audio & Media

- **Voice-Acted NPC Audio** — Captain Salita has dedicated intro, success, and fail clips per island
- **Ingay Voice Clips** for the per-island taunting/warning moments
- **Per-Island Background Music** with the option to override the default Mixkit track
- **Expo Audio Playback Engine** with a dedicated audio Zustand store for global playback control
- **Mute Button & Volume Control** accessible across the app
- **Cloudinary-Hosted Audio Pipeline** for challenge passages, explanations, and NPC clips

Teacher Mode

- **Role-Based Teacher Portal** for accounts with the TEACHER role
- **Student Roster Dashboard** listing every assigned student
- **Performance Analytics** showing per-island accuracy, completion rate, and progress timeline
- **Mistake Review** highlighting the specific challenges and choices students got wrong
- **Badge Visibility** so teachers can see milestone progress across the class

Admin Mode

- **Audio Upload Console** for ADMIN role accounts to upload and manage challenge audio
- **Cloudinary-Backed Asset Management** with delete-audio and upload-audio endpoints
- **Content-Only Access** keeping admins focused on media curation, separate from teacher analytics

Onboarding

- **Animated Login & Welcome Flow** with Google OAuth and traditional username/password options
- **Story-Driven Onboarding** introducing Dagat, Captain Salita, and the threat of Ingay

- **Step-by-Step Tutorial** teaching learners the map, single-play audio rule, hints, and badges

UI/UX Design

Design Principles

- **Pirate-Adventure Theming:** Ocean-deep navy, gold accents, and weathered-map aesthetics throughout
- **Child-Friendly Interface:** Bright, encouraging visuals with large, touch-friendly buttons
- **Accessibility First:** Large tap targets, readable typography, character-mode toggle for learners who prefer a quieter UI
- **Responsive Layout:** Adapts across phone and tablet form factors
- **Smooth Animations:** React Native Reanimated 4 powered transitions on every screen

Theme System

- **NativeWind (Tailwind) Theming:** Consistent ocean / gold / parchment design tokens
- **Per-Island Visual Identity:** Each island carries its own palette, music, and shard art
- **Custom Typography:** Optimized for young readers with strong contrast on deep-ocean backgrounds
- **Animated Characters & Scenes:** Captain Salita, Dagat, and Ingay each have dedicated animated components

Security & Privacy

Data Protection

- **JWT Authentication** via Jose with secure token storage in Expo SecureStore
- **Google OAuth Integration** through @react-native-google-signin/google-signin and Google Auth Library on the backend
- **Role-Gated APIs:** STUDENT, TEACHER, and ADMIN roles enforced at every Next.js route
- **Server-Side Answer Validation** — correct answers never ship to the client at challenge time, preventing client-side cheating
- **HTTPS-Only API Endpoints** deployed on Vercel

Privacy Features

- **Minimal Data Collection:** Only username, email, and progress data are stored
- **No Audio Recording from Learners:** LinguaQuest is listen-and-answer; no microphone access is requested
- **Scoped Multiplayer Visibility:** Chat messages are room-scoped and tied to active sessions only
- **Permission Transparency:** All required permissions declared in app.json

Performance Optimizations

Efficiency Features

- **TanStack React Query Caching** for islands, pins, progress, and badges to minimize network calls
- **Zustand Stores** for lightweight, fast global state across auth, audio, and multiplayer
- **Pusher Real-Time Channels** instead of polling for multiplayer state synchronization
- **Lazy-Loaded Island Assets:** Background music and shard art load only when an island is entered
- **Cloudinary CDN Delivery** for fast audio streaming across regions

Resource Management

- **Audio Player Lifecycle Cleanup:** Expo Audio resources released when leaving challenge screens
- **Animation Lifecycle Handling:** Reanimated values properly disposed across navigation transitions
- **Connection-Aware Behavior:** @react-native-community/netinfo detects offline states gracefully
- **Prisma Connection Pooling** via Neon serverless PostgreSQL for elastic backend scaling

Supported Devices

Minimum Requirements

- **Android Version:** 7.0 (API level 24)
- **iOS Version:** 14.0+

- **RAM:** 2GB minimum, 3GB recommended
- **Storage:** ~150MB for app and cached audio assets
- **Network:** Stable internet for audio streaming and multiplayer

Optimal Experience

- **Android Version:** 11.0+ (API level 30+)
- **iOS Version:** 16.0+
- **RAM:** 3GB or more
- **Display:** 5.5" or larger screen
- **Audio:** Headphones strongly recommended for listening accuracy
- **Network:** Wi-Fi or stable 4G/5G for smooth multiplayer

Note: LinguaQuest can run on Expo Go for quick previews, but the production build is distributed as a development/EAS build to enable Google Sign-In and SecureStore on real devices.

Acknowledgments

Technologies Used

Frontend

- **Expo SDK 55 / React Native 0.83 / React 19** — Cross-platform runtime
- **expo-router** — File-based, typed navigation
- **NativeWind (Tailwind)** — Utility-first styling
- **Zustand** — Lightweight global state for auth, audio, and multiplayer
- **TanStack React Query** — Server state and caching
- **Expo Audio** — Audio playback for passages and NPC dialogue
- **React Native Reanimated 4** — Declarative animations
- **Pusher JS** — Real-time multiplayer channels
- **Expo SecureStore** — Encrypted token storage
- **react-native-view-shot** — Certificate export from the profile screen
- **@react-native-google-signin/google-signin** — Google OAuth on device

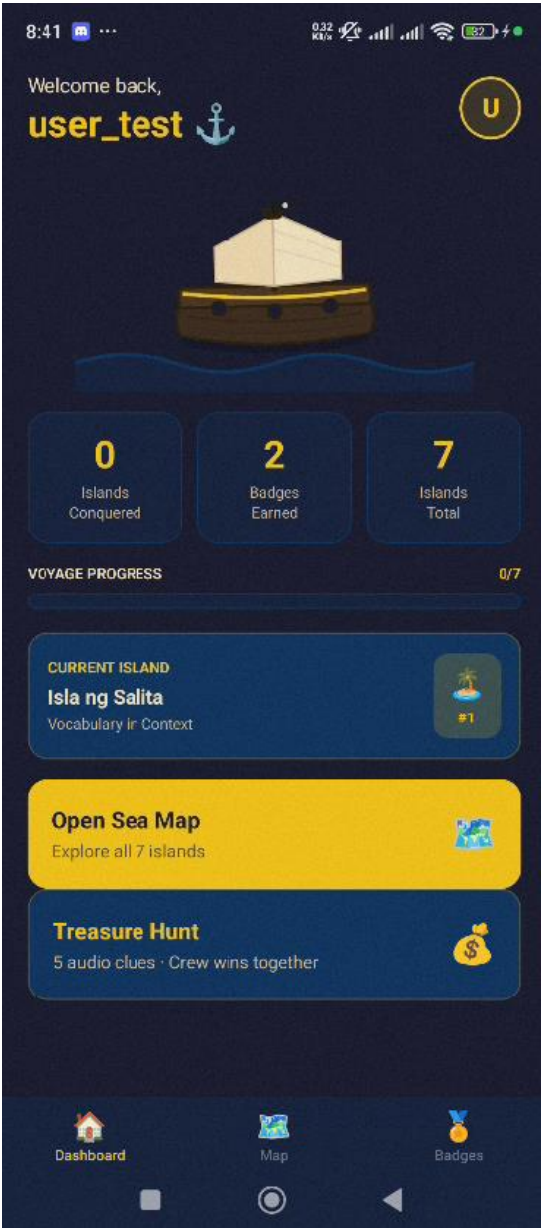
Backend

- **Next.js 16** — API routes deployed to Vercel
- **Prisma ORM** — Type-safe database access
- **Neon Serverless PostgreSQL** — Elastic, serverless database
- **Jose** — JWT signing and verification
- **Pusher** — Server-side real-time event broadcasting
- **Cloudinary** — Audio and image asset hosting
- **Google Auth Library** — Server-side Google ID token verification

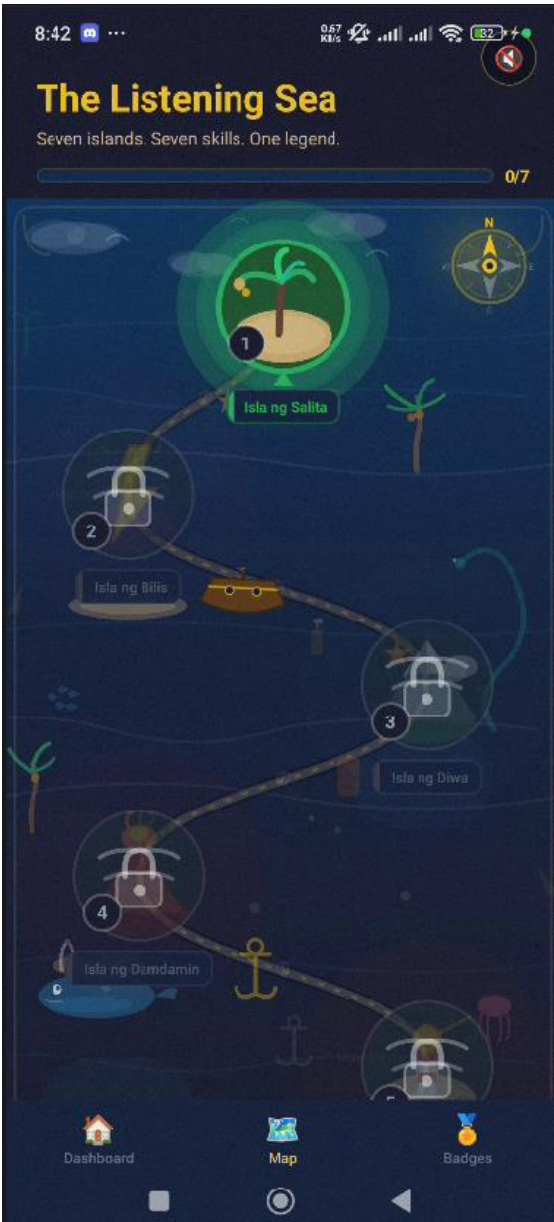
Educational Content

- Seven-island story arc and listening objectives crafted around Grade 7 Filipino comprehension milestones
- Audio passages, questions, hints, and explanations authored to scaffold attempts and reward correct listening
- Multiplayer challenge bank curated to reinforce mastery in collaborative crew battles
- Story frame designed around Filipino-rooted characters (Dagat, Captain Salita, Ingay) and Tagalog island names

PAGE OVERVIEW



HOME



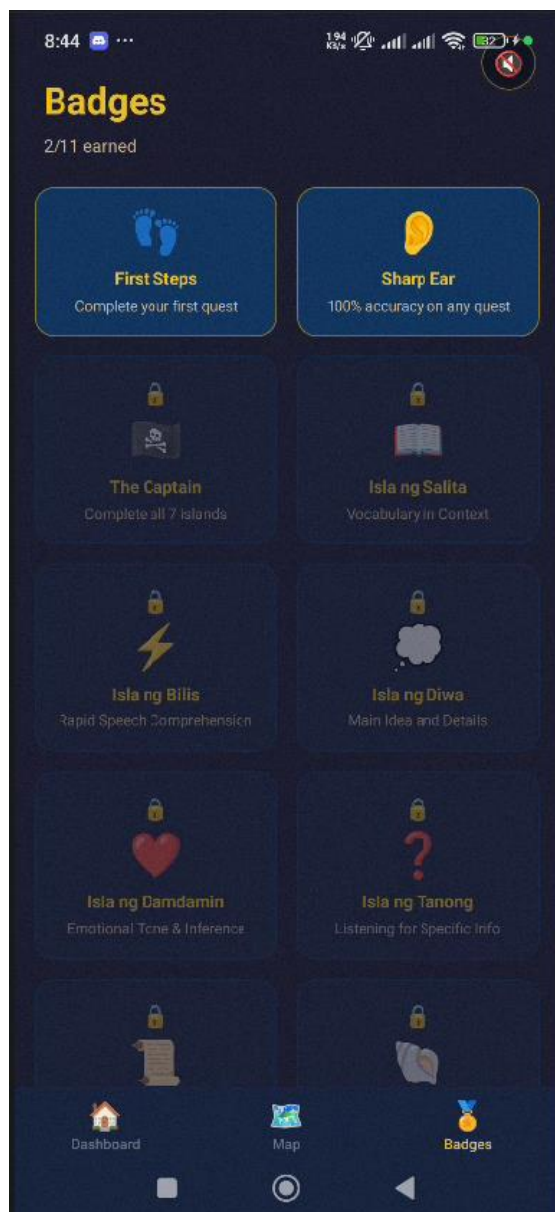
MAP



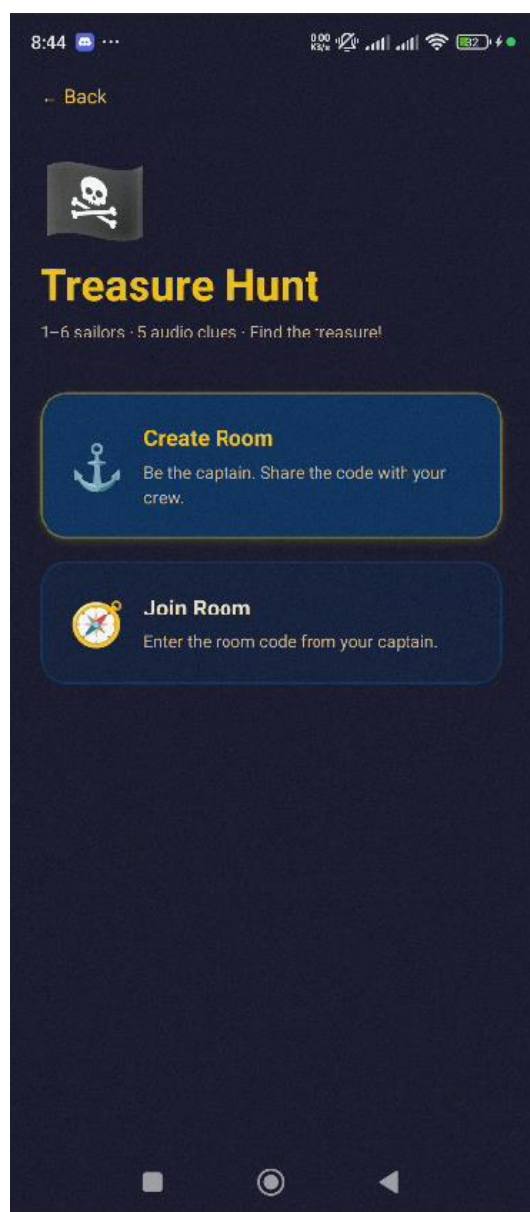
INSIDE THE MAP



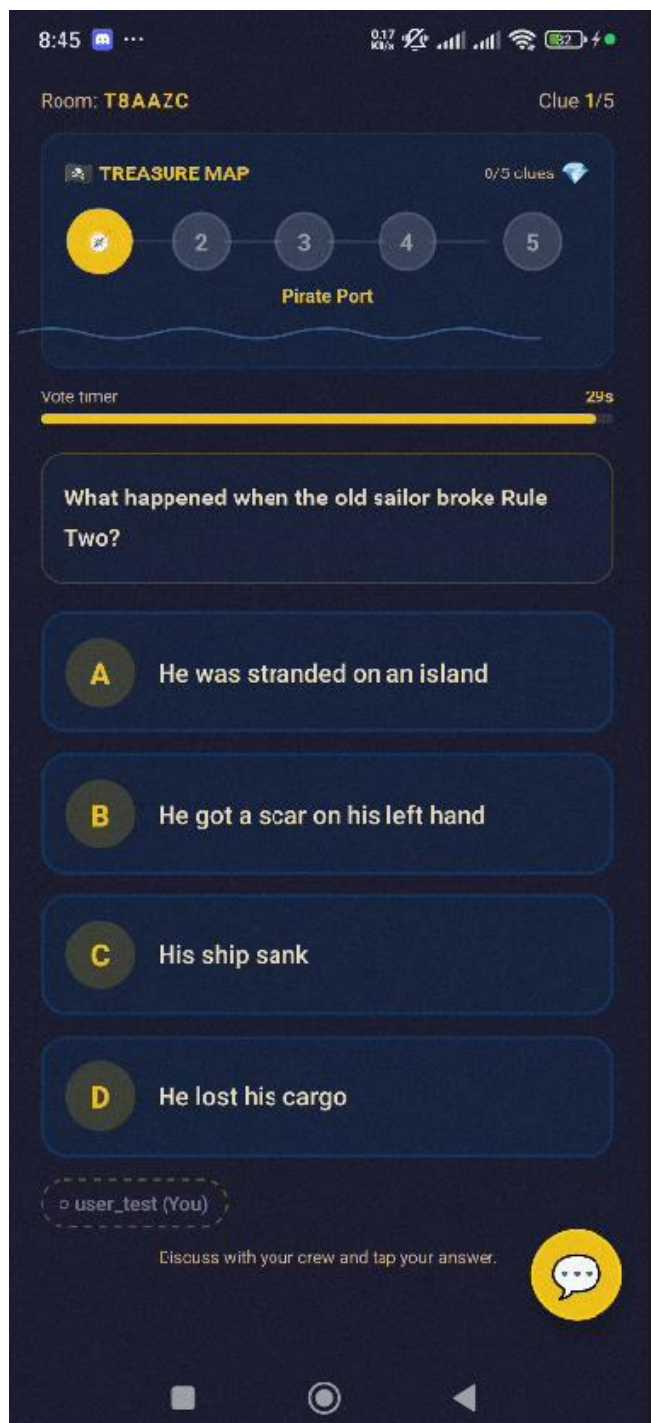
THE QUEST



BADGES



MULTIPLAYER MODE



INSIDE MULTIPLAYER